Journal Pre-proof

An innovative design of bionic anti-reflux biliary stent with conical spiral valve based on functionally graded materials

Shuxian Zheng, Shikun Cai, Xiaofei Song, Jian Qi, Jinheng Yan

PII: S1000-9345(25)00156-7

DOI: https://doi.org/10.1016/j.cjme.2025.100152

Reference: CJME100152

To appear in: *Chinese Journal of Mechanical Engineering*

Received date: 1 August 2024
Revised date: 10 October 2025
Accepted date: 7 November 2025

Please cite this article as: Shuxian Zheng, Shikun Cai, Xiaofei Song, Jian Qi and Jinheng Yan, An innovative design of bionic anti-reflux biliary stent with conical spiral valve based on functionally graded materials, *Chinese Journal of Mechanical Engineering,* (2025) doi:https://doi.org/10.1016/j.cjme.2025.100152

This is a PDF of an article that has undergone enhancements after acceptance, such as the addition of a cover page and metadata, and formatting for readability. This version will undergo additional copyediting, typesetting and review before it is published in its final form. As such, this version is no longer the Accepted Manuscript, but it is not yet the definitive Version of Record; we are providing this early version to give early visibility of the article. Please note that Elsevier's sharing policy for the Published Journal Article applies to this version, see: https://www.elsevier.com/about/policies-and-standards/sharing#4-published-journal-article. Please also note that, during the production process, errors may be discovered which could affect the content, and all legal disclaimers that apply to the journal pertain.

© 2025 The Authors. Publishing services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd.

An innovative design of bionic anti-reflux biliary stent with conical spiral valve based on functionally graded materials

Shuxian Zheng[1], Shikun Cai[1], Xiaofei Song[1,*], Jian Qi[2], Jinheng Yan[1]

Author Details

[1], Tianjin, 300354, China. [2]Tianjin Key Laboratory of High Speed Cutting and Precision Machining, Tianjin High-end Intelligent CNC Machine Tool Engineering Research Center, Tianjin University of Technology and EduTianjin Key Laboratory of Equipment Design and Manufacturing Technology, School of Mechanical Engineering, Tianjin Universitycation, Tianjin, 300222, China.

Abstract

Biliary stenting is an important treatment for malignant biliary obstruction; however, the procedure can cause complications, such as duodenal biliary reflux. In this paper, a novel conical spiral valve anti-reflux biliary stent based on functionally graded material (FGM) was proposed. This bionic design, inspired by the annular epidermal microstructure of leeches, innovatively achieves adaptive deformation-straightening under antegrade flow to reduce resistance and shrinking under reflux to block retrograde flow. To optimize the valve's performance, a FGM gradient was designed with a power function (n=2) for shear modulus distribution, allowing the valve to balance high stiffness at the large end (to resist fluid impact) and low stiffness at the small end (to achieve tight closure under low reflux pressure). Additionally, the valve model was subjected to 2D equivalent fluid-structure interaction (FSI), and finite element layering approach was used to design and optimize the physical parameters of the graded material. Simulation results showed that the FGM valve with the conical spiral structure achieved an axial closure rate of 90.45%, a fluid passage rate of 0.76%, and a radial deformation angle of 6.28°. Performance evaluations indicated that compared with homogeneous valves, the FGM valve improved anti-reflux performance by 79.27% while enhancing forward drainage. In vitro fluid experiments further validated these findings, confirming the significant superiority of the FGM valve in anti-reflux performance. The gradient material design provides valuable insights for the development of medical anti-reflux valves, highlighting its potential to advance clinical solutions for malignant biliary obstruction.

Keywords: Biliary stent, Functionally graded material, Fluid-Structure Interaction, Bionic design

Journal Pre-proof

1 Introduction

Biliary tract malignant tumors are common digestive system tumors worldwide [1]. In addition to bile duct and gallbladder cancers that originate in the biliary system, cancers such as pancreatic cancer and hepatocellular carcinoma are prone to invade the bile ducts and cause malignant biliary obstruction, which leads to persistent obstructive jaundice, which not only impairs the patient's quality of life, but may also lead to serious complications such as liver and kidney failure to death, surgical resection is the only curative treatment. However, most patients with multiple liver metastases, extensive tumor invasion of bile ducts and blood vessels, and other unfavorable factors cannot tolerate surgery; only a few patients are suitable for surgical resection[2-4]. Therefore, biliary stenting is currently the preferred palliative treatment for malignant biliary obstruction[5]. This procedure was first performed by Soehendra et al. using a plastic stent, which was simple to place but prone to blockage in a relatively short time[6,7]. As research progressed, self-expanding metallic stents (SEMS) were developed with longer drainage patency, which were placed across the papilla and can impair the function of the sphincter of Oddi, leading to biliary sludge or food residue blockage, which in turn results in failure of the SEMS blockage [8,9]. In the reported relevant cases, the incidence of cholangitis after SEMS placement ranged from 6.5% to 22%, and almost all patients demonstrated reflux problems [10], which seriously affects the quality of the patient's survival [11].

*Correspondence: xiaofeisong@tju.edu.cn
[1] Tianjin Key Laboratory of Equipment Design and Manufacturing Technology, School of Mechanical Engineering, Tianjin University, Tianjin, 300354, China
Full list of author information is available at the end of the article

To address the problem of biliary reflux after implantation, Hu applied anti-reflux metallic stents (ARMS) to patients with biliary malignancies, and designed a hemispherical silicone valve with a cross-shaped outlet at the end of the metallic stent, which demonstrated that it was feasible to apply anti-reflux biliary stents in the clinic, even though malfunctions due to tumor growth and dislocation of the stent were seen. Lee designed an S-type anti-reflux valve using silicone material, and no reflux cholangitis with acute pancreatitis occurred during the follow-up period, although stent occlusions were found in one-third of the patients [12]. Hamada designed an anti-reflux stent with a funnel-like structure with an approximately 10 mm valve, which had a longer patency time but a higher rate of migration [13]; a year later, they optimized the design, which had a significant advantage over SEMS stents in overcoming food embedding while maintaining the ability to resist migration [14]. Kin designed a duckbill-shaped anti-reflux metal stent with the valve designed at the end of the metal stent, and showed that it could be used more effectively in patients with malignant biliary obstruction [15]. In summary, the existing design cannot take into account the problems of smooth drainage, reduction of stent displacement and anti-reflux, and it is necessary to design a novel type of biliary anti-reflux valve stent to overcome the existing defect.

Leeches are cold-blooded annelid whose skin has evolved a tightly arranged ring-like structure [16], as shown in Fig. 1a. When subjected to tensile loading, the ring-like structure gradually separates and the protrusion decreases until it is completely straightened, as shown in Fig. 1b, at this time, the fusiform body shape reduces its resistance to fluid, making it easy to attach to rocks in the pool or on moving animals; the original structure can also be restored after the load is removed. Based on the principle of bionics, this paper envisions a retractable bionic valve structure, as shown in Fig. 1c, which is less resistant to fluid and has a lower chance of displacement in downstream flow, shrinks and closes in reflux, providing an anti-reflux effect. This design not only meets the needs of anti-reflux valves under different flow directions, but also ensures the functionality and durability of the biliary stent.

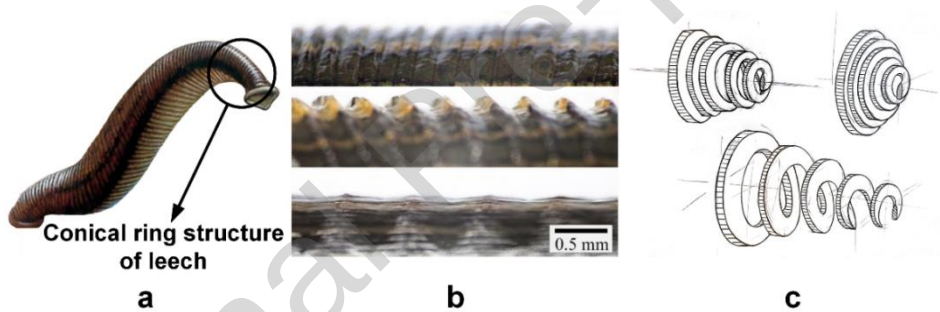

Figure 1 (a) Annelid leech, (b) Epidermal microstructure of leech, (c) Structural conception diagram of bionic conical spiral valve

In order to achieve better performance, the use of functionally graded materials (FGM) for the manufacture of anti-reflux valves was conceived for the gradient-varying structure of the conical spiral. FGM was first proposed by Koizumi et al. [17], referring to the continuous change in the composition and structure of a material in space, which results in a continuous change in the material's properties[18]. For the anti-reflux valve characteristics, which need to meet the requirements of guided downflow and regurgitant closure, FGM was applied to the conical spiral valve to achieve a gradient of elastic modulus from high to low, from the large end to the small end, i.e. the higher elastic modulus at the large end ensures the valve remains stable and open in downflow, while keeping the morphology less distorted when subjected to fluid washout; the modulus of elasticity of the conical spiral valve varies layer by layer up to the small end, where the lower modulus of elasticity allows closure of the valve under small regurgitant pressures, preventing duodenal fluid from entering the bile duct and causing inflammation.

For the study of gradient materials in mechanics, there are analytical methods [19,20], finite element layering method [21,22], isotropic gradient cell method [23] and meshless analysis method [24,25]. Among them, the finite element layering method [26-28]simulates the properties of a gradient material by layering the target structure along the gradient direction of the material and assigning different parameters to each layer; the accuracy is positively correlated with the number of layers, which is superior for the gradient structure analysis of valves. In addition, the use of fluid-structure interaction (FSI) can well reflect the dynamic changes in the structural and fluid domains and can simulate the environment of the biliary tract and analyze the mechanical properties of the valve. Currently, there are few studies on FSI for anti-reflux valves. The time-consuming nature of FSI at the 3D level is not conducive to experimenting with various gradient ratios of materials for

Journal Pre-proof

optimal design. In this regard, this paper utilizes a multilayered 2D model for equivalent simulation of the 3D model, which ensures the accuracy of the simulation, and at the same time, produces the results in a faster manner. Further, the results of the 2D simulation are input into the 3D model for FSI, which can be used to verify and evaluate the effect.

Based on bionic design principles, this paper proposes a novel anti-reflux biliary stent featuring a conical spiral valve fabricated from functionally graded materials (FGM). Unlike existing anti-reflux stents, such as hemispherical or duckbill-shaped designs, that struggle to balance smooth drainage, anti-reflux efficacy, and anti-migration performance, the proposed bionic conical spiral valve, inspired by the annular structures of leeches, inherently adapts to flow directions: it reduces migration risk during antegrade flow by minimizing resistance and enhances closure during reflux to block retrograde flow. To further optimize performance, FGM with a shear modulus varying continuously via a power function was innovatively incorporated, resolving the long-standing conflict between the need for large-end stability (to withstand fluid impact) and small-end flexibility (to enable tight closure). Additionally, a 2D equivalent fluid-structure interaction (FSI) method, an improvement over time-consuming 3D simulations, was developed to efficiently optimize gradient parameters, combined with a finite element layering approach for material analysis. This integrated design (structure - material - method) overcomes key limitations of conventional stents. Through the 2D equivalent FSI and finite element layering methods, the FGM valve's properties were analyzed and optimized, with its anti-reflux and forward drainage capabilities evaluated. Finally, a fluid test bench was also constructed to simulate in vivo biliary tract conditions, while the FGM anti-reflux valve was fabricated and its design effectiveness validated via in vitro experiments.

2 Material and methods

2.1 Anatomy of the Human Upper-limb

The mainstream method for palliative treatment of malignant biliary obstruction (Fig. 2a) is to implant a biliary stent to allow smooth drainage of bile. Compared with plastic stents, metal stents are strong, corrosion-resistant, and have a long service life. Based on the current status of research, a compressible, multi-grouped, wavy metal stent was used as a dilatation stent. As the implantation of a biliary stent makes the sphincter of Oddi dysfunctional and easily causes reflux cholangitis, which affects the quality of patient's survival [10], an anti-reflux valve is set at the end of the stent close to the duodenum; the valve is able to respond passively by utilizing the flow of bile (Fig. 2b, c). Furthermore, in conjunction with the FGM design, the modulus of elasticity of the valve material (silicone) is gradient along the axis to reduce the possibility of aberration failure while meeting the requirements of anti-reflux.

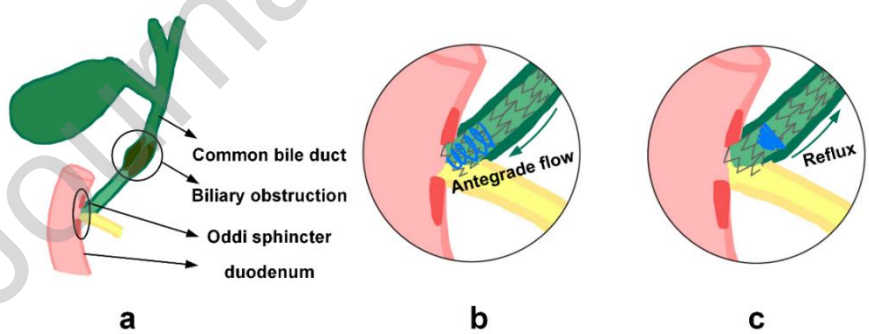

Figure 2 (a) Cancer causing biliary obstruction, (b) Antegrade flow after stent implantation, (c) Resistance to reflux after stent implantation

2.1.1 Metal stent

The average diameter of the common bile duct in healthy adults is about 6 mm [29]. Considering the clinical studies of existing biliary stents and drainage requirements, the metal stent was designed to have an outer diameter of 10 mm and an inner diameter of 9.4 mm when fully dilated, made of nickel-titanium alloy. The structure of the metal stent, as shown in Fig.

Journal Pre-proof

3a, consists of multiple spaced wavy metal wires distributed in the axial direction. It can be inserted into the bile duct under compression and is supportive and pliable after dilatation. The length of the common bile duct averages about 60 mm, so the total length of the metal stent is lengthened or shortened according to the imaging results.

2.1.2 Conical spiral anti-reflux valve

The conical spiral anti-reflux valve was set on the duodenal side within the stent, and the structure is shown in Fig. 3b. Based on the inner diameter of the stent, the outer diameter was $D = 9.4$ mm. The overlap rate O between each layer of the valve is determined by $O = (1 - \Delta D/(2 \times W)) \times 100\%$. D is the difference of outer diameter between each layer of the valve, when $O = 50\%$ represents the midline of the upper cross-section and the inner diameter of the lower cross-section are co-linear. Too low an overlap will lead to structural failure of the valve when subjected to reflux, so $O = 60\%$ was set, the valve is determined to be 5-turn by the overlap rate and the outer diameter of the stent. The conical spiral anti-reflux valve was proposed to be manufactured from silicone. After preliminary experiments, the cross-sectional thickness $T = 0.6$ mm and width $W = 1.9$ mm of each layer were set to facilitate material gradient design. On this basis, the layer spacing P of the valve was set to 2.5 times the cross-sectional thickness of a single layer at 1.4 mm to ensure smooth bile drainage when there is no pressure difference in the entero-biliary environment.

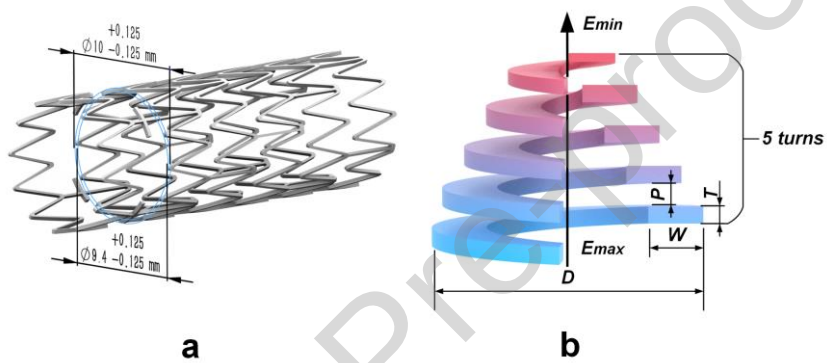

Figure 3 (a) Design of metal stent, (b) Design of conical spiral valve

2.2 Simulation methods

2.2.1 Establishment of FSI model

The performance of conical spiral anti-reflux valves is determined by a combination of structural design and material properties. Here, an FSI model is developed to experiment with multiple gradient ratios of valve materials. FSI is a coupled research method for the structural and fluid domains of the model, which is time-consuming and inefficient in a single pass for a complex 3D model that undergoes deformation. Therefore, 2D equivalent FSI was used to reflect the mechanical properties of the 3D valve accurately and efficiently [30].

To carry out the simulation of a 2D valve, simulation nodes should be selected first. Since the valve structure is a 5-turns conical spiral, the ratio of turns to simulation sections can be selected as 1:1, 1:2, 1:3, etc., in which the 2D equivalence of 1:1 is too rough, and a ratio of 1:3 or higher will lead to a larger amount of work in error calibration, which will offset the high efficiency of 2D simulation, so the ratio of 1:2 is set. An equally spaced selection of nodes allows for better representation of flow field variations in the simulation and facilitates the interpolation of FGM. So, the valve was dissected along the XOZ plane to and 1~11 Layers can be obtained, as shown in Fig. 4(a,b). Sections 1~11 representing the mechanical properties of the respective layers were selected, as shown in Figure 4c.

In the simulation process, in addition to the simulation nodes, it is also necessary to establish the mechanical connection between the nodes. The ontological models established for silicone usually use Ogden's exponential model [31] and *N*th polynomial model, and the *N*th polynomial model includes the Mooney-Rivlin model [32], the Neo-Hookean model [33], and the Yeoh model [34]. Since this study intends to describe the intrinsic relationship of the material by changing mechanical parameters of the material, the *N*th order polynomial model was chosen, considering the number of parameters and the property of FGM, the Mooney-Rivlin model was chosen as the intrinsic model of silicone material [33]. Since the Poisson's ratio of silicone is close to 0.5, it is an incompressible superelastic material [35]. In this case, the functional expression for

Journal Pre-proof

the energy density W of the strain energy is given by:

$$W = C_{10}\left(\bar{I}_1 - 3\right) + C_{01}\left(\bar{I}_2 - 3\right) \tag{1}$$

$$G_0 = 2\left(C_{10} + C_{01}\right) \tag{2}$$

where C_{10} is associated with C_{01}, both describing temperature-dependent constants of the material [36], $C_{10} = 152.5$ MPa and $C_{01} = 19.6$ MPa; $\bar{I}_1$ and $\bar{I}_2$ are the first and second invariants of the deformed gradient tensor, respectively; G_0 is the initial shear modulus of the silicone, substituting C_{10} and C_{01}, $G_0 = 344.2$ MPa is calculated, and the mechanical properties of FGM in the axial direction vary based on G_0.

Since the two-dimensional equivalent model was established to study the axial tensile compression deformation characteristics of the valve, the radial section was not considered; therefore, the 11 sections were simplified into rigid models; with the density and Young's modulus of the profile defined as those of rigid material. To accurately simulate the axial non-linear deformation, non-linear springs *k* were added between adjacent sections. Additionally, for movement stability, a linear damping *c*=0.1 was connected in parallel with each spring, with values based on empirical data. Sections 5 and 6 were selected as examples to establish a mechanical model, as shown in Fig. 4d. Considering the large end of the model, Layer 1-3 were cast and fixed to the stent during valve fabrication, while Layer 3-11 were able to move in response to the fluid forces. Non-linear springs *k* were respectively established between sections 3-11, with initial stiffness values of the spring $k_3^0 \sim k_{11}^0$ measured via simulation tests on the corresponding valve layers.

After initial spring stiffness values were measured, the 2D model was calibrated for displacement error to ensure equivalence of axial mechanical properties between 2D and 3D models. A generic static analysis step was employed for both models with an initial time step of 5×10^{-5} s. Self-contact properties were configured with a friction coefficient μ=0.2, determined from empirical data of multiple simulation experiments, and contact control was established. The same axial load was applied to both models. Taking Sections 5 and 6 as example, the midpoints of the outer sides of each section were designated as calibration points. Axial coordinates were iteratively calibrated until the error fell below 5%, ultimately calibrating spring stiffnesses k_3-k_{11} that met the requirements.

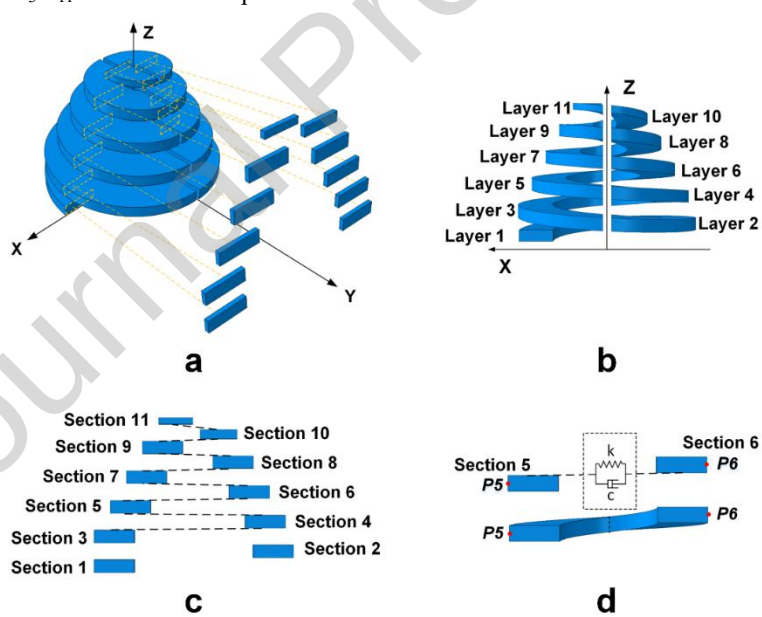

Figure 4 (a) Association between Section and Layer, (b) Each half turn of valve divided into a Layer, (c) Assignment of mechanical simulation links between Sections, (d) Mechanical simulation model

2.2.2 Principles for selecting material gradients

The performance of the valve plays a key role in the anti-reflux effect of the entire stent. After simulation modeling of the structure, FSI analysis is required to find the most suitable material ratio for the structure, and the anti-reflux performance of the valve is judged by two indicators: axial closure rate φ and radial deformation angle θ.

Axial closure rate φ characterize the valve's ability to resist regurgitation. A higher φ indicates tighter inter-layer fit and superior anti-reflux performance, while a lower φ fails to meet anti-reflux requirements. The outer center point *P* (shown in

Journal Pre-proof

Fig. 4d) of sections 1-11 serves as the reference point. Gap measurements between sections are illustrated in Fig. 5a, with the φ calculation formula defined as follows.

$$\varphi = \frac{1}{8} \times \sum_{i=2}^{9} \varphi_i = \frac{1}{8} \times \sum_{i=2}^{9} (1 - \frac{z_{i+2} - z_i - 0.6}{1.4 - 0.6}) \tag{3}$$

where φ_i represents the closure rate between the ith reference point and the neighboring point, φ denotes the arithmetic mean of φ_i. The fluid passage rate is defined as the ratio of the flow rate of the pipeline with the valve to that without the valve under identical loading conditions. It is evident that a higher axial closure rate corresponds to a lower fluid passage rate of the reflux. As illustrated in Fig. 5b, the relationship between the axial closure rate and fluid passage rate indicates that when the axial closure rate exceeds 85%, the flow rate of the reflux falls below 1% of the overall flow rate, thereby satisfying the performance requirements.

Radial deformation angle θ serves as an index that characterizes a valve's capacity to withstand radial deformation. Following FEA and preliminary experimental investigations, it was determined that the difficulty of radial deformation varies by direction, with the most significant deformation occurring along the negative X direction. Consequently, the radial deformation angle θ is defined as the arctangent of the ratio of the displacement caused by gravitational load to the original outer diameter. As illustrated in Fig. 5c, the four reference points p_5, p_7, p_9, p_{11} located along the positive X semiaxis, exhibit considerable displacements and are utilized for calculating θ_5, θ_7, θ_9 and θ_{11}. The formula is as follows:

$$\theta = \tan^{-1}\left(\frac{1}{4}\sum_{i=2}^{5}\theta_{2i+1}\right) = \tan^{-1}\left(\frac{1}{4}\sum_{i=2}^{5}\frac{\Delta x_{2i+1}}{x_{2i+1}}\right) \tag{4}$$

θ_{2i+1} represents the degree of radial deformation at each reference point, θ denotes the arithmetic mean of θ_{2i+1}. Following the FEA, it was concluded that the valve is susceptible to deformation and a loss of anti-reflux function when $\theta \geq 10^\circ$.

Thus, the FGM of the valve is chosen with constraints of a radial deformation angle $\theta \leq 10^\circ$ and an axial closure rate $\varphi \geq 85\%$, while maximizing the axial closure rate φ as the optimization objective. By employing a joint simulation method utilizing Abaqus and X-flow, various gradients of FGM were simulated to identify the most effective configuration.

In contrast to homogeneous materials where components are uniformly distributed in space, the physical parameters of functionally FGMs vary along the coordinate axis. This variation is what makes FGMs advantageous over homogeneous materials in certain application scenarios.

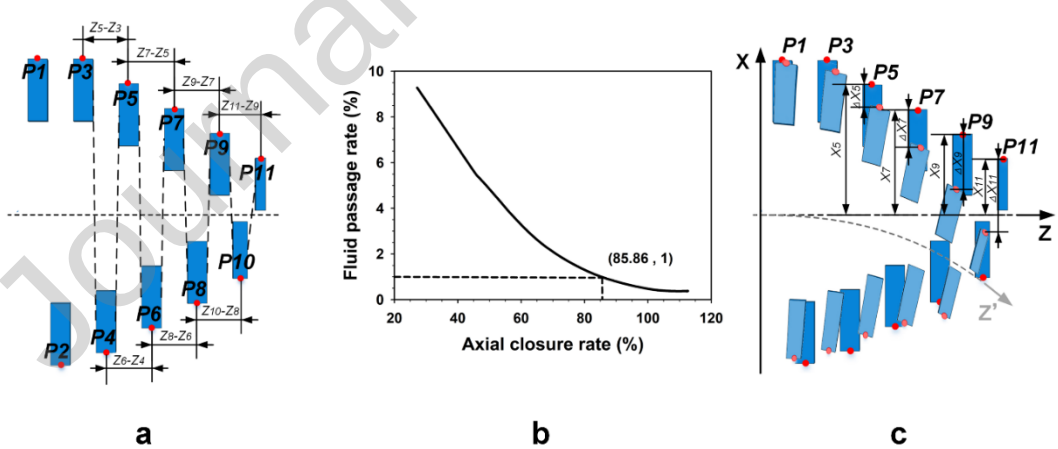

Figure 5 (a) Axial closure rate of the valve, (b) Axial closure rate vs. fluid passage rate, (c) Radial deformation angle of the valve

2.2.3 Determination the shear modulus of FGM

To design the valve material using the FGM method, it is essential to confine the gradient variation within specified boundaries. The larger end must exhibit higher stiffness and a greater shear modulus to minimize distortion when subjected to fluid impact. Conversely, the smaller end should possess increased flexibility and a reduced shear modulus to enhance the axial closure rate under lower reflux pressure. Consequently, it is imperative to first ascertain the shear modulus at both the top and bottom ends of the valve. The initial shear modulus of the silicone material was calculated to be 344.2 MPa from Eq. (1). To encompass a broad spectrum of shear moduli, this value was scaled up and down using a 2-fold logarithmic scale,

Journal Pre-proof

as detailed in Table 1.

Table 1 ten sets of shear moduli in logarithmic scale

Serial No.	G_1	G_2	G_3	G_4	G_5	G_6	G_7	G_8	G_9	G_{10}
Shear modulus/MPa	688.4	344.2	172.1	86.05	43.03	21.51	10.76	5.378	2.689	1.345
C_{10}	305	152.5	76.25	38.13	19.06	9.531	4.766	2.383	1.191	0.596
C_{01}	39.2	19.6	9.8	4.9	2.45	1.225	0.613	0.306	0.153	0.077

By inputting these 10 sets of experimental data into the simulation model of a homogeneous valve, the axial closure rate and radial deformation angle varying with shear moduli can be obtained. For the axial compression experiments conducted on each model group, a pressure of 2.29 KPa was applied at the onset of reflux, The criteria for the inital selection of the shear modulus at the bottom and top end were established based on an axial closure of $\varphi_2 > 95\%$ at the reference point P_2 at the large end and $\varphi_9 > 95\%$ at the reference point P_9 at the small end.

2.2.4 Evaluation of anti-reflux biliary stent

The anti-reflux performance of the FGM valve was assessed through FSI simulations conducted on a 3D model. The structural domains of both the FGM valve featuring an optimal material gradient and the homogeneous valve were constructed using Abaqus 2018. In this context, the shear modulus of the homogeneous valve was taken as G_3; the fluid domain was modeled in X-flow.

The supportive and flexible properties of the nickel-titanium metal stent were simulated and evaluated using Abaqus 2018 [37]. The simulation parameters are shown in Table 2 [38,39]. The friction coefficient $\mu=0.2$, or both the stent's self-contact and the contact between the valve and stent, was established based on empirical data derived from multiple simulation experiments. The stent was discretized into 43,184 linear hexahedral cells (C3D8I), while the valve comprised 214,800 linear hexahedral cells (C3D8H) [40]. The stent's support was assessed by applying radial pressure, and its flexibility was evaluated through angular displacement applied to the ends of the stent.

Table 2 simulation parameters of nickel-titanium shape memory alloy

parameter	ρ	E_A	ν_A	E_M	ν_M	ε_L	$(\delta\sigma/\delta T)_L$
value	6450 kg/m^3	68000 MPa	0.3	36000 MPa	0.3	0.04	6.5
parameter	σ_L^S	σ_L^E	T_0	$(\delta\sigma/\delta T)_U$	σ_U^S	σ_U^E	σ_{CL}^S
value	520 MPa	635 MPa	37 ℃	6.5	180 MPa	26 MPa	780 MPa

2.3 In vitro experimental validation

Related studies on anti-reflux valves primarily utilize silicone or PTFE [41,42]. Given the gradient structural properties of conical spiral valves, this study employed a two-component additively molded liquid silicone as the fabrication material. The long vulcanization time associated with thermosetting materials, such as silicone, complicates direct 3D printing. Therefore, an indirect 3D printing process was utilized to create molds, allowing for the preparation of the functional gradient valve through a casting process.

2.3.1 Mold design

Based on the structural characteristics of the valve, a casting mold was designed, as illustrated in Fig. 6a, featuring a "slideway" groove shaped as a conical helix. The outer edges at the top and bottom were specifically designed to obstruct the flow of silicone liquid, thereby minimizing the difficulty of demolding. The draft angle was set at 6°, the overall mold thickness was 0.7 mm, and the wall thickness was 0.6 mm, with the mold pitch matching that of the valve. Resin molds were produced using light-curing 3D printing.

2.3.2 Casting process

1) Pre-curing time

For medical silicone materials, pre-curing time denotes the duration required for the A and B components to achieve the desired pouring viscosity following their mixing. During this period, the viscosity of the silicone increased progressively over time. Preliminary experiments indicate that the optimal casting viscosity of silicone is about 12000 MPa·s. The

Journal Pre-proof

fabrication of the FGM valve necessitates various silicone materials for the joint casting process. Given the significant variability in performance associated with different pre-curing times, this duration is established through experimental methods [43].

Hardness serves as the primary parameter indicating the performance of silicone. Based on simulations, three silicone materials with hardness values of 5° Shore A, 15° Shore A, and 40° Shore A were selected for the fabrication of the FGM valve. Fig. 6b illustrates the relationship between pre-curing time and viscosity at 20° C. Consequently, the blending order was established as 5° →15° →40° Shore A.

2) Curing temperature

High-temperature curing is essential for optimizing the properties of silicone valves, as this process significantly influences the durability of valves constructed from silicone materials. Therefore, an experiment was designed to examine the mechanical properties of silicone materials at various curing temperatures. Tensile tests were conducted on 2 mm thick specimens of 40° Shore A silicone, which were cured at different temperatures for 4 hours, as illustrated in Fig. 6c. The tensile strength curve of the specimens, depicted in Fig. 6d, demonstrates a decrease in tensile strength from 2.38 MPa to 1.88 MPa as the vulcanization temperature rises from 30° C to 90° C. Additionally, Fig. 6e presents the elongation at break curve, indicating that the elongation at break of the material diminished from 260.41% to 225.07% with the increase in curing temperature from 30° C to 90° C. The tensile experiments revealed that 40° Shore A silicone exhibited the greatest sensitivity to curing temperature regarding both tensile strength and elongation at break (Fig. 6d-e), underscoring its importance for valve durability. Although 5° and 15° Shore A silicones were incorporated into the FGM gradient, their temperature dependence was secondary to the optimized gradient function (n=2), which inherently alleviated localized material weaknesses. Future research will extend the validation of curing temperature across all hardness levels.

The overall results show that an increase in curing temperature leads to a reduction in both the tensile strength and elongation at break of silicon, thereby diminishing the material properties. Consequently, maintaining the curing temperature at 30° C during the preparation process yields improved mechanical properties.

2.3.3 Valve preparation

Based on simulations, mold design and the casting process, two types of valves were fabricated. Three silicone variants with Shore A hardness values of 5° , 15° , and 40° were selected for the joint casting of FGM valves. The homogeneous control valve was constructed using 40° Shore A silicone, which serves as a clinically established benchmark for anti-reflux applications. Although the lower hardness silicones (5° and 15° Shore A) demonstrate higher elongation rates, simulation results (Table 1, Fig. 8a) revealed their susceptibility to radial deformation (θ>12°), making them unsuitable as standalone materials. After the sequence of pre-curing, pouring silicone liquid, static curing, demolding, and residual silicone removal, the final homogeneous valve and FGM valve are illustrated in Fig. 6f.

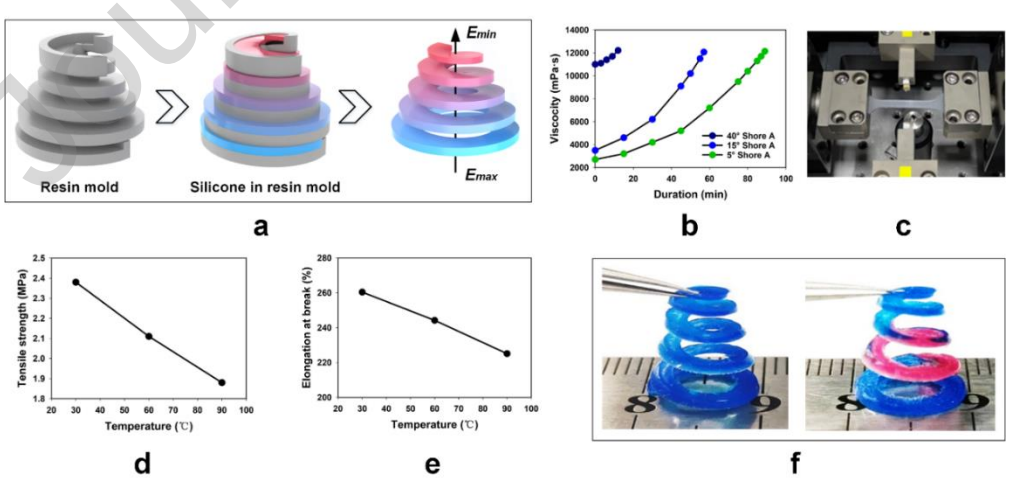

Figure 6 (a) Fabrication of valve mold, (b) Pre-curing process of three types of silicone, (c) Tensile testing of silicone, (d) Tensile strength curve under temperature change, (e) Elongation at break curves under temperature change, (f) Product of homogeneous valve and FGM valve

Journal Pre-proof

2.3.4 In vitro experimental platform construction

After fabrication of the valve, a flow-pressure control system was designed to validate the antegrade drainage and anti-reflux ability of the FGM valve. System components are shown in Fig. 7a. This system simulates the physiological environment inside the bile duct, such as the length, internal diameter, bile duct pressure, etc. Among them, the volumetric cylinder uses the siphon effect to output liquid; the peristaltic pump has the function of flow adjustment, with the Robert clamp to realize the stability of the liquid level of the volumetric cylinder, so the pressure of water flow is stable and adjustable; the flowmeter monitored the line flow; two micro-pressure gauges detected the line pressure in experimental and control group. Fig. 7b shows the valve in the simulated bile duct. The pipe was made of transparent resin material, and by changing the orientation of the valve in the bile duct, both antegrade and reflux experiments can be simulated.

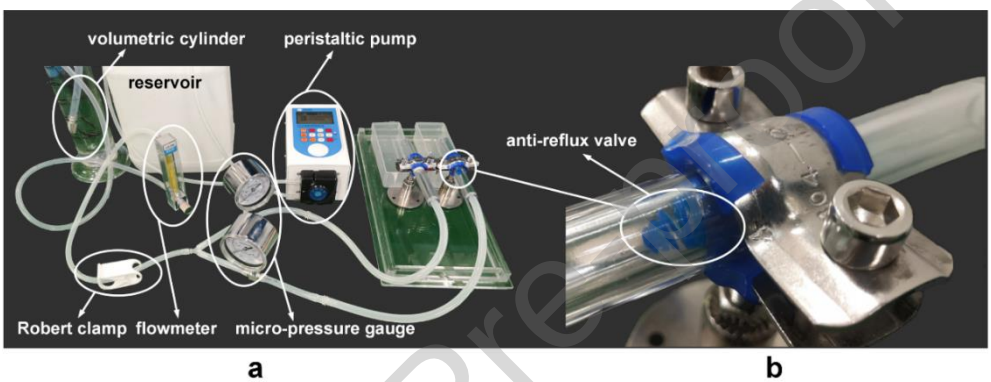

Figure 7 (a) Construction of in vitro experimental platform, (b) Anti-reflux valve in simulated bile duct

Journal Pre-proof

3 Results

3.1 Simulation result of valve material gradient

Based on the experimental design in Section 2.2.3, the axial closure rate curves for $G_3 \sim G_{10}$ V homogeneous valves were derived as shown in Figure 8a. G_1 and G_2 were excluded due to poor closure rate. According to the above simulations, some of the reference points in the figure have a closure rate greater than 100%, which is due to the deformation and tilt of the soft homogeneous valve in the axial compression simulation. According to the above simulations, the shear modulus at the bottom section of the valve G_{bottom} was set to G_3, and the shear modulus at the top section G_{top} was set to G_{10}.

After determining the material properties at the two ends of the FGM, at present, the way of selecting the physical parameters of each layer of FGM mainly divided into two categories: one introduces the concept of volume fraction, i.e., according to volume fractions of different phase materials to derive the parameters at different positions of FGM; the other category defines the parameters in the direction of the grading directly as a Power function or an Exponential function. In this paper, we use the volume fraction-based Voigt model to describe the distribution pattern of physical parameters of graded materials along the axial [44-46]. Based on the shear modulus of the top and bottom ends of the valve, we constructed gradient functions with multiple trends of the interlayer material as simulation objects[47,48].

The Voigt model is utilized to describe the distribution of the physical parameters of a graded material along the axial direction, essentially assigning the parameters of the material to the gradient in terms of the weights of the constituent phases, i.e.

$$P = \sum_{i=1} P_i V_i \tag{5}$$

$$\sum_{i=1} V_i = 1 \tag{6}$$

where P_i denotes the physical parameters (e.g., density, elastic modulus, etc.) of the i th constituent phase material, V_i denotes the volume fraction of the i th phase material. This model is relatively simple and applicable; many scholars' current research on FGM is based on this model [49,50].

The material gradient established based on Exponential function is represented as follows:

$$G(i) = G_b V_b(i) + G_t V_t(i), i = 1, 2, ..., 11 \quad (7)$$

$$G(1) = G_b = G_3; G(11) = G_t = G_{10} \quad (8)$$

where i is the material layer of the valve divided along the axial. The parameters α, β and G_i in Eq. (7) can be solved by combining Eq. (8). In this way, the shear modulus of each simulation node of the valve can be obtained under the Exponential function.

When establishing the gradient of a material based on Power function, a linear mixture model is introduced as follows:

$$G(i) = G_b V_b(i) + G_t V_t(i), i = 1, 2, ..., 11 \quad (9)$$

$$V_b(i) = ((11 - i)/10)^n, i = 1, 2, ..., 11 \quad (10)$$

$$V_t(i) = 1 - V_b(i) \quad (11)$$

To simplify the simulation process, taking n to a total of five values of 0.2, 0.5, 1, 2 and 5, which represent five different gradient trends of Power functions.

As shown in Fig. 8b, the gradient shear modulus under the trend of five Power functions and one Exponential function. It can be seen that in FGM, the gradient under the Power function relationship, the larger Power index n leads to a smaller overall shear modulus of the valve; the gradient under exponential relationship, the overall shear modulus of the valve is only higher than that of the Power function with $n = 5$.

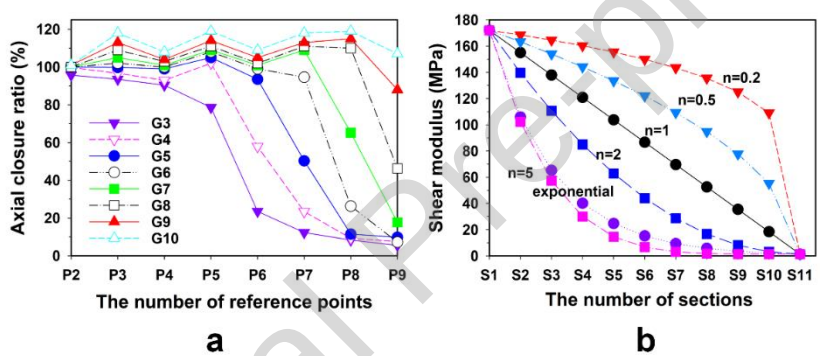

Figure 8 (a) Axial closure rate at each reference point in $G_3 \sim G_{10}$, (b) Six sets of shear modulus with different gradients

2D FSI simulations were conducted on the above six valves with different gradient materials, as shown in Fig. 9a-f, which is the flow field diagram of the valve under a reflux pressure of 2.29 kPa. Red represents a higher fluid flow rate, which slows down gradually toward blue. Among them, Fig. 9a shows the valve at $n = 0.2$ for the power function, which is obtained from Eq. (3), with an axial closure of 54.4442% and a fluid passage rate of 4.11%; Fig. 9b is the power function for $n = 0.5$, with axial closure rate 62.5355% and fluid passage rate 2.83%; Fig. 9c is the power function for $n = 1$, with axial closure rate 69.8248% and fluid passage rate 2.10%; Fig. 9d shows the power function for $n = 2$, with axial closure rate 90.4520% and fluid passage rate 0.76%; Fig. 9e shows the power function for $n = 5$, with axial closure rate 99.7803% and fluid passage rate 0.51%; Fig. 9f shows the valve for exponential function, with axial closure rate 98.0330%, and fluid passage rate 0.53%.

As shown in Fig. 9g, the following pattern can be derived: the greater the n of the Power function, the greater the axial closure rate of the valve and the greater the radial deformation angle. The FGM with Exponential function and Power function of $n = 5$ are excluded due to not satisfy the condition of radial deformation angle (i.e., lower than the red dashed line in Fig. 9g), the Power function of $n = 2$ has the optimal fluid passage rate and satisfies the condition, so this set of gradient functions is selected as the design parameter for the FGM of valve. With the corresponding axial closure rate of 90.45%, fluid passage rate of 0.76% and radial deformation angle of 6.28°.

Journal Pre-proof

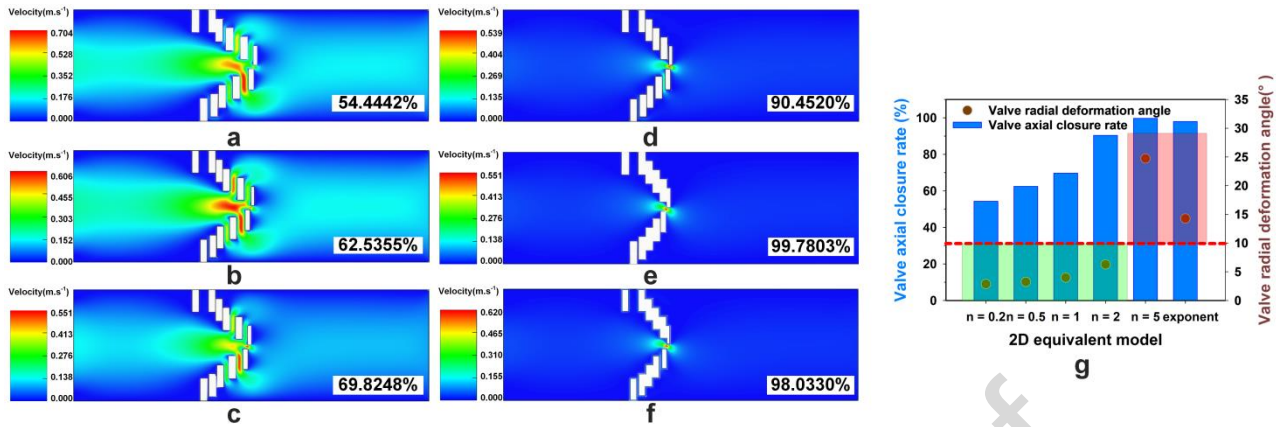

Figure 9 Flow field diagrams of valves against reflux of **(a)** power function with n = 0.2, **(b)** power function with n = 0.5, **(c)** power function with n = 1, **(d)** power function with n = 2, **(e)** power function with n = 5, **(f)** exponential function; **(g)** Optimization of axial closure rate and radial deformation angle for six sets of gradient function

3.2 Evaluation of anti-reflux biliary stent with FGM valve

Fig. 10(a, b) shows the flow field diagrams of the FGM valve and homogeneous valve at a reflux pressure of 2.29 kPa. Combined with the axial closure ratio at each reference point in Fig. 10(c), the FGM valve showed improved results over the homogeneous valve at each reference point, especially at the small end P_6-P_9 of the valve. At this point, the fluid passage rate was 0.9720% for the FGM valve and 4.6881% for the homogeneous valve; the anti-reflux performance of the FGM valve was improved by 79.27% compared with the control group. Fig. 10(d, e) shows the flow field diagrams of the FGM valve and the homogeneous valve at an antegrade pressure of 2.29 kPa. Combined with the axial elongation at each reference point in Fig. 10(f), the axial elongation of the FGM valve is higher than that of the homogeneous valve at each reference point. At this time, the axial elongation of the FGM valve was 201.93%, and 93.77% for the homogeneous valve; the axial elongation of the FGM valve increased by 1.15 times compared with the control group.

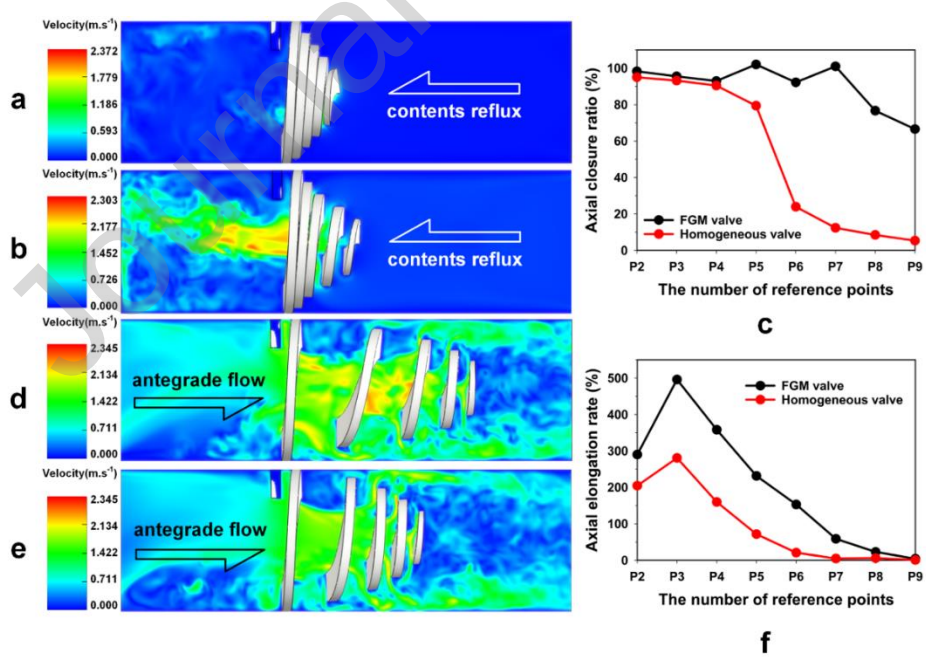

Figure 10 Anti-reflux flow field diagram of **(a)** FGM valve, **(b)** Homogeneous valve; **(c)** Axial closure ratio of two valves in anti-reflux;

Journal Pre-proof

Antegrade flow field diagram of **(d)** FGM valve, **(e)** Homogeneous valve; **(f)** Axial elongation ratio of two valves in antegrade flow

The anti-reflux biliary stent has a different structure at the valve-stent junction than conventional stents, which requires a radial mechanical analysis of the affected metal stent [51]. Fig. 11a-c shows the assessment of the supportability of the FGM valve stent, homogeneous valve stent and conventional stent subjected to radial pressure. As can be seen from the figures, the presence of the valve increases the radial support stiffness of the stent. Fig. 11d shows the change in mean diameter for all three at 0-30 kPa radial pressure. The mean diameter is defined as the average of the diameters at each inflection point of the support ring; the trend suggests that the valve increases the radial support stiffness and independent of the valve type. Bending simulations were performed to determine the flexibility of the stent, as shown in Fig. 11e-g. Since the valve is placed inside the stent near the biliary end of the duodenum, the stents with and without the valve have similar bending stiffness, as shown in Fig. 11h, which does not adversely affect the flexibility.

3.3 In vitro results

Fig. 12a shows the flow trends of the two valves in the simulated bile ducts at several downstream pressures between 1.29 and 3.29 kPa; the difference in flow was small, proving that the presence of the FGM valve has a small effect on the patency of the bile duct. The reflux flow rates of the two valves in the simulated bile ducts under different reflux pressures are shown in Fig. 12b.When the reflux occurs, pressure difference between the duodenum and the bile duct is 2.29-4.25 kPa [52,53], taking eight sets of countercurrent pressures for experiment, FGM valve subjected to reflux of 2.29-4.25 kPa with a reflux flow rate less than 8%, and the effect becomes better with increasing reflux pressure.

Journal Pre-proof

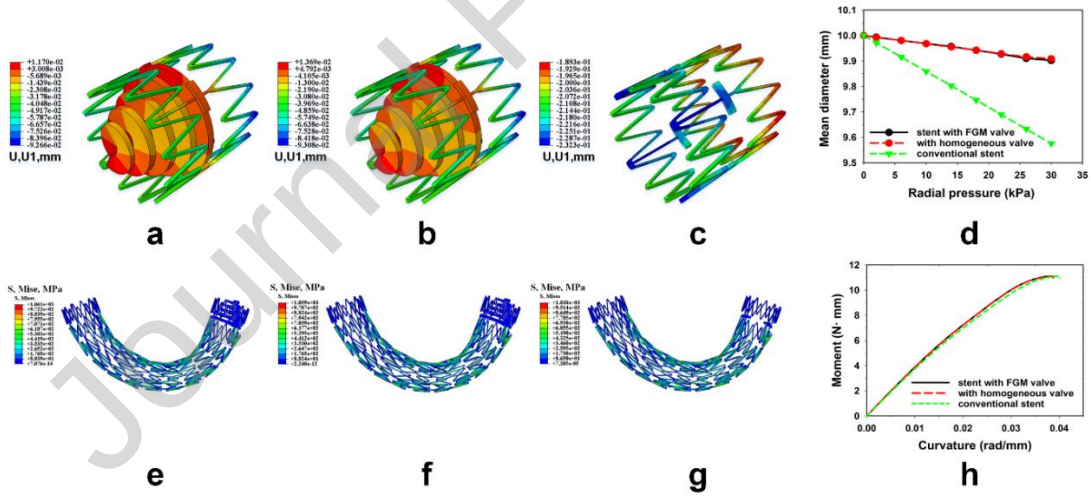

Figure 11 Von mises stress distribution under radial pressure of (a) Stent with FGM valve, (b) Stent with homogeneous valve, (c) Conventional stent; (d) Mean diameter variation of the three types of stents under radial pressure; Von mises stress distribution under bending of (e) Stent with FGM valve, (f) Stent with homogeneous valve, (g) Conventional stent; (h) Curvature vs. moment for three types of stents under bending

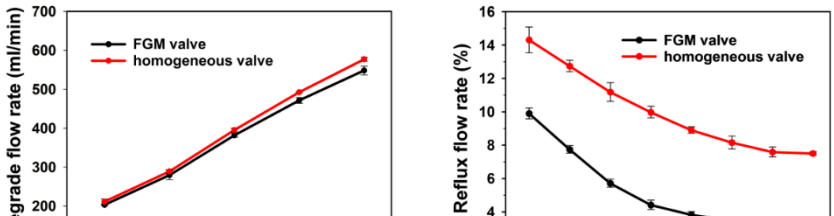

Figure 12 **(a)** Comparison of two valves antegrade drainage effects, **(b)** Comparison of the anti-reflux effect of two valves

4 Discussion

In this paper, a 2D FSI simulation method was presented to accelerate the model simulation, which is able to achieve higher simulation accuracy at a faster speed after calibration; at the same time, by applying the finite element layering method, under the premise of satisfying the axial mechanical properties, more accurate gradient function characteristics of FGM can be accurately fed back.

As shown in Fig. 9g, the following pattern can be derived: a larger n of the Power function (the smaller the overall modulus of elasticity of the valve) not only leads to a better closure performance, but also results in a larger radial deformation angle, potentially causing aberration or failure. Under simulation conditions, the gradient function of the material expressed as a Power function of n=2 is optimal, the axial closure rate reaches 90.45%, the fluid passage rate is 0.76%, and the radial deformation angle is 6.28°, all of which collectively meet the operational requirements.

In section 3.2, the simulation evaluation showed that during anti-reflux, the small end of the FGM valve fits more tightly than the homogeneous valve to reduce reflux flow; and in antegrade flow, it is able to elongate as a whole, allowing for the smooth drainage of bile. Comparing the support and flexibility of FGM valve stent, homogeneous valve stent and conventional stent, the support of stent with FGM valve was improved compared with the conventional stent; at the same time, the impact on flexibility was small and did not affect the safety and feasibility of stent implantation. The design was able to reduce discomfort for patients with implanted stents.

In vitro experiments were conducted by casting silicone through 3D printed molds in order to fabricate FGM valves and homogeneous valves that meet the design requirements. The results show that the FGM valve exhibits significant advantages in both anti-reflux and drainage performance.

In anti-reflux experiments, the fluid passage rate of the FGM valve was less than 8% at pressures ranging from 2 to 4 kPa, and decreased from 7.8% to 3.1% with increasing reverse pressure, which resulted in a combined improvement in anti-reflux performance of 50% compared with that of the homogeneous valve at different pressures. In the downstream experiments, the difference in flow between the bile ducts installed with FGM valves and unloaded bile ducts at different pressures was small, indicating that the FGM valves basically did not affect the downstream patency of the bile ducts, and the FGM valves were able to achieve good and smooth drainage. Therefore, the design of the FGM valve can reduce the regurgitation of biliary contents, effectively reduce the risk of valve displacement, and reduce the incidence of cholangitis.

It has been reported that the effectiveness of anti-reflux biliary stents is still controversial; the effectiveness mainly depends on the structure and material of the anti-reflux valve. Displacement, collapse, and folding are the main causes of valve dysfunction and stent obstruction. Clinically, there is no ideal anti-reflux biliary stent that has been comprehensively promoted [54,55]. The novel anti-reflux biliary stent proposed herein is an open-ended stent with good patency and anti-reflux properties compared with similar stents, and has the potential to relieve biliary obstruction and solve the reflux induced by stent implantation.

The core innovations of this work lie in four aspects: (1) Bionic structural design. The conical spiral valve, inspired by leech epidermis, achieves adaptive deformation for both drainage and anti-reflux, outperforming fixed-shape valves (e.g., funnel-like, S-type). (2) FGM application. The power function-based gradient (n=2) of shear modulus balances the mechanical requirements of different valve segments, a breakthrough compared to homogeneous materials. (3) Efficient

Journal Pre-proof

simulation. The 2D equivalent FSI method, calibrated with 3D results, significantly reduces optimization time while ensuring accuracy, facilitating iterative design. (4) Fabrication and validation. Indirect 3D printing of molds for multi-hardness silicone casting (5°, 15°, 40° Shore A) enables practical realization, with in vitro experiments confirming superior performance under physiological conditions. These innovations provide a new paradigm for medical anti-reflux valve design.

The FGM valve's fluid passage rate of 0.76% under reflux conditions surpasses existing designs, including duckbill (5.2%, [15]) and funnel-shaped valves (3.8%, [12]), by leveraging a biomechanically optimized gradient and bionic geometry. This advancement addresses the critical clinical need for durable anti-reflux performance without compromising drainage efficiency. This study also has some limitations. For example, the effect of the pH of the reflux fluid on the service life of the silicone valve was not explored. In addition, the safety of stent implantation should be considered. Subsequent studies will focus on clinical studies of stents and potential complications.

5 Conclusions

In response to the therapeutic needs of bile reflux in the clinic and the limitations of the current research on anti-reflux biliary stent valves, we proposed an innovative bionic anti-reflux biliary stent with conical spiral valve based on FGM. A 2D FSI method was utilized to simulate and optimize the ratios of FGM to better meet the needs of use; an *in vitro* experimental platform was also constructed to validate the FGM valve. The simulation and *in vitro* experimental results show that this design can better meet the requirements of clinical use.

(1) To address clinical needs for mitigating bile reflux and overcome limitations of existing anti-reflux biliary stent valves, this study proposed an innovative bionic anti-reflux biliary stent featuring a conical spiral valve based on functionally graded materials (FGM).
(2) A 2D fluid-structure interaction (FSI) method was employed to simulate and optimize the FGM ratios, ensuring the design better aligns with clinical application requirements.
(3) An in vitro experimental platform was constructed to validate the FGM valve. Both simulation and in vitro results confirmed that the proposed design meets clinical performance demands effectively.

Future work will advance in vivo animal studies to assess the FGM valve stent's long-term biocompatibility, tissue response, and durability. More complex gradient models (e.g., combining exponential and power functions) will be explored to better fit the biliary biomechanics. Additionally, homogeneous valves with 5°, 15°, and 40° Shore A silicone will be fabricated for comparative experiments, quantifying how hardness impacts key performance metrics.

Authors' Contributions

Shuxian Zheng and Xiaofei Song were in charge of the whole trial; Shicun Cai and Jinhe wrote the manuscrip; Jian Qi assisted with sampling and laboratory analyses. All authors read and approved the final manuscript.

Authors' Information

Shuxian Zheng is currently an associate professor at *School of Mechanical Engineering, Tianjin University, China*. She received her PhD degree from *Xi'an Jiao Tong University, China*, in 2007. Her research mainly focuses on additive manufacturing and bio-manufacturing. E-mail: sxzheng@tju.edu.cn

Shikun Cai born in 1998, received the master degree from *Tianjin University, China*, in 2005. He did research works on additive manufacturing and bio-manufacturing. E-mail: ziiswan@126.com

Xiaofei Song is currently an associate professor *at School of Mechanical Engineering, Tianjin University, China*. She received her Ph.D. degree from *Tianjin University, China*, in 2008. Her research focuses on ultrasonic machining and medical device. E-mail: xiaofeisong@tju.edu.cn

Qi Jian is currently a lecturer at *School of Mechanical Engineering, Tianjin University of Technology and Education, China*. He received his PhD degree from *Tianjin University, China*, in 2019, and his main research direction is additive manufacturing process research. E-mail: qijian78@tju.edu.cn

Jinheng Yan born in 1998, received the master degree from *Tianjin University, China*, in 2004. He did research works on bioengineering and medical device. E-mail: yanjinheng180@163.com

Journal Pre-proof

Competing Interests

The authors declare no competing financial interests.

Acknowledgements

Not applicable

.

Funding

Supported by National Natural Science Foundation of China (Grant No. 52175275, 52275457)

References

[1] M F Y Viesca, M AHurvanitakis. Early diagnosis and management of malignant distal biliary obstruction: a review on current recommendations and guidelines. *Clinical And Experimental Gastroenterology*, 2019, 12: 415–432. https://doi.org/10.2147/CEG.S195714

[2] R A Dutra, J S dos Santos, W M de Araújo, et al. Evaluation of hepatobiliary excretion and enterobiliary reflux in rats with biliary obstruction submitted to bilioduodenal or biliojejunal anastomosis. *Digestive Diseases and Sciences*, 2008,53(4): 1138–1145. https://doi.org/10.1007/s10620-007-9955-5

[3] O Shah, P Shah, S Zargar. Hepaticocholecystoduodenostomy compared with Roux-en-y choledochojejunostomy for decompression of the biliary tract. *Annals Of Saudi Medicine*, 2009, 29(5): 383–387. https://doi.org/10.4103/0256-4947.55169

[4] A J Grossberg, L C Chu,C R Deig, et al. Multidisciplinary standards of care and recent progress in pancreatic ductal adenocarcinoma. *Ca-a Cancer Journal for Clinicians*, 2020, 70(5): 375–403. https://doi.org/10.3322/caac.21626

[5] T H Lee, J H Moon, S H Park. Biliary stenting for hilar malignant biliary obstruction. *Digestive Endoscopy*, 2020, 32(2), 275–286. https://doi.org/https://doi.org/10.1111/den.13549

[6] N Soehendra, V Reynders-Frederix. Palliative bile duct drainage - a new endoscopic method of introducing a transpapillary drain. *Endoscopy*, 1980, 12(1): 8–11. https://doi.org/10.1055/s-2007-1021702

[7] U Seitz, H Vadeyar, N Soehendra. Prolonged patency with a new-design Teflon biliary prosthesis. *Endoscopy*, 1994, 26(5): 478–482. https://doi.org/10.1055/s-2007-1009008

[8] D N Reddy, R Banerjee, O W Choung. Antireflux biliary stents: are they the solution to stent occlusions? *Current Gastroenterology Reports*, 2006, 8(2): 156–160. https://doi.org/10.1007/s11894-006-0012-x

[9] S Doi, Y Namura, T Kikuyama, et al. Novel anti-reflux biliary metal stent with a distal tapered end for distal malignant biliary obstruction: a feasibility study.*Endoscopy International Open*, 2023, 11(05): E460–E465. https://doi.org/10.1055/a-2039-3853

[10]B Hu, T T Wang, Z M Shi, et al. A novel antireflux metal stent for the palliation of biliary malignancies: a pilot feasibility study. *Gastrointestinal Endoscopy*, 2011, 73(1): 143–148. https://doi.org/10.1016/j.gie.2010.08.048

[11]S P Misra, M Dwivedi, Reflux of duodenal contents and cholangitis in patients undergoing self-expanding metal stent placement. *Gastrointestinal Endoscopy*, 2009, 70(2): 317–321. https://doi.org/10.1016/j.gie.2008.12.054

[12]K J Lee, M J Chung, J Y Park, et al. Clinical advantages of a metal stent with an S-shaped anti-reflux valve in malignant biliary obstruction. *Digestive Endoscopy*, 2013, 25(3): 308–312. https://doi.org/10.1111/j.1443-1661.2012.01382.x

[13]T Hamada, H Isayama, Y Nakai, et al. Novel antireflux covered metal stent for recurrent occlusion of biliary metal stents: A pilot study. *Digestive Endoscopy*, 2014, 26(2): 264–269. https://doi.org/10.1111/den.12112

[14] T Hamada, H Isayama, Y Nakai, et al. Antireflux metal stent with an antimigration system for distal malignant biliary obstruction: a feasibility pilot study. *Surgical Laparoscopy Endoscopy & Percutaneous Techniques*, 2015, 25(3): 212–217. https://doi.org/10.1097/SLE.0000000000000149

[15] T Kin, K Ishii, Y Okabe,et al. Feasibility of biliary stenting to distal malignant biliary obstruction using a novel designed metal stent with duckbill-shaped anti-reflux valve. *Digestive endoscopy*, 2021, *33*(4): 648–655. https://doi.org/10.1111/den.13827

[16] L Pulze, N Baranzini, R Girardello, et al. A new cellular type in invertebrates: first evidence of telocytes in leech, Hirudo Medicinalis. Scientific Reports, 2017, 7: 13580. https://doi.org/10.1038/s41598-017-13202-9

[17]M Koizumi, M Niino. Overview of FGM research in Japan. *MRS Bulletin*, 1995, *20*: 19–21. https://api.semanticscholar.org/CorpusID:137735819

[18]S Alkunte, I Fidan, V Naikwadi, et al. Advancements and Challenges in Additively Manufactured Functionally Graded Materials: A Comprehensive Review. *Journal of manufacturing and Materials Processing*, 2024, *8*(1), 23. https://doi.org/10.3390/jmmp8010023

[19] A Mahi, E A Bedia, A Tounsi. A new hyperbolic shear deformation theory for bending and free vibration analysis of isotropic, functionally graded, sandwich and laminated composite plates. *Applied Mathematical Modelling*, 2015, 39(9): 2489–2508. https://doi.org/10.1016/j.apm.2014.10.045

[20] W Xu, G Pan, Z Moradi, et al. Nonlinear forced vibration analysis of functionally graded non-uniform cylindrical microbeams applying the semi-analytical solution. *Composite Structures*, 2021, 275. https://doi.org/10.1016/j.compstruct.2021.114395

[21] Y S Wang, D Gross. Analysis of a crack in a functionally gradient interface layer under static and dynamic loading. In W. Hwang & K. S. Han (Eds.), *Fracture And Strength Of Solids, Pts 1 And* 2, 2000, 183-187: 331~336. https://doi.org/10.4028/www.scientific.net/KEM.200,183-187.331

[22] Y S Wang, G Y Huang, D Gross. On the mechanical modeling of functionally graded interfacial zone with a Griffith crack: Anti-plane deformation. *Journal Of Applied Mechanics-Transactions Of The Asme*, 2004, 125 (1-2) : 189-205. https://doi.org/10.1115/1.1641066

[23] S Suresh, A Mortensen, *Fundamentals of functionally graded materials*. 1998,https://api.semanticscholar.org/CorpusID:136502900

[24] J G Wang, G R Liu. A point interpolation meshless method based on radial basis functions. *International Journal For Numerical Methods In Engineering*, 2002, 54(11): 1623–1648. https://doi.org/10.1002/nme.489

Journal Pre-proof

[25]X G Li, B D Dai, L H Wang. A moving Kriging interpolation-based boundary node method for two-dimensional potential problems. *Chinese Physics B*, 2010, 19(12). https://doi.org/10.1088/1674-1056/19/12/120202

[26] L L Wei, H L Leo, Q Chen. Structural and hemodynamic analyses of different stent structures in curved and stenotic coronary artery. *Frontiers in Bioengineering and Biotechnology*, 2019, 7. https://doi.org/10.3389/fbioe.2019.00366

[27] H Yaakobovich, R P Mayo, U Zaretsky, et al. Numerical models of valve-in-valve implantation: effect of intentional leaflet laceration on the anchorage. *Biomechanics and Modeling in Mechanobiology*, 2020, 19(2), 415–426. https://doi.org/10.1007/s10237-019-01218-1

[28] H P Xue, S C Saha, S Beier, et al. Topological optimization of auxetic coronary stents considering hemodynamics. *Frontiers In Bioengineering And Biotechnology*, 2021, 9. https://doi.org/10.3389/fbioe.2021.728914

[29]S H Ahn, C An, S S Kim, et al. CT evaluation of long-term changes in common bile duct diameter after cholecystectomy. *Journal of the Korean Society of Radiology*, 2023, 85(3): 581–595. https://doi.org/10.3348/jksr.2023.0031

[30] I Fumagalli, L Dede', A Quarteroni, A. A reduced 3D-0D fluid–structure interaction model of the aortic valve that includes leaflet curvature. *Biomechanics and Modeling in Mechanobiology*, 2025, 24(4): 1169–1189. https://doi.org/10.1007/s10237-025-01960-9

[31] Y Yamashita, H Uematsu,S Tanoue. Calculation of strain energy density function using ogden model and mooney-rivlin model based on biaxial elongation experiments of silicone rubber. *Polymers*, 2023, 5(10): 2266. https://doi.org/10.3390/polym15102266

[32]S Hosseini, G Rahimi, G. Experimental and numerical analysis of hyperelastic plates using Mooney-Rivlin strain energy function and meshless collocation method. *Engineering Analysis with Boundary Elements*, 2023, 150: 199–218. https://doi.org/10.1016/j.enganabound.2023.02.024

[33] B Kim, S B Lee, J Lee, et al. A Comparison Among Neo-Hookean Model, Mooney-Rivlin Model, and Ogden Model for Chloroprene Rubber. *International Journal of Precision Engineering and Manufacturing*, 2012, 13(5): 759–764. https://doi.org/10.1007/s12541-012-0099-y

[34] Q P Xu, J Y Liu. An improved dynamic model for a silicone material beam with large deformation. *Acta Mechanica Sinica*, 2018, 34(4): 744–753. https://doi.org/10.1007/s10409-018-0759-y

[35]E Porte, N Pashine,S K Patiballa, et al. Variable-stiffness metamaterials with switchable Poisson's ratio. *Device*, 2025, 3(1): 100570. https://doi.org/https://doi.org/10.1016/j.device.2024.100570

[36]A Avanzini, Battini, D. Integrated experimental and numerical comparison of different approaches for planar biaxial testing of a hyperelastic material. *Advances in Materials Science and Engineering*, 2016, 2016(1):6014129. https://doi.org/10.1155/2016/6014129

[37]N S Ribeiro, J Folgado,H C Rodrigues. Surrogate-based visualization and sensitivity analysis of coronary stent performance: A study on the influence of geometric design. *International Journal For Numerical Methods In Biomedical Engineering*, 2018, 34(10). https://doi.org/10.1002/cnm.3125

[38]M Azaouzi, A Makradi, S Belouettar. Deployment of a self-expanding stent inside an artery: A finite element analysis. *Materials & Design*, 2012, 41(10): 410–420. https://doi.org/10.1016/j.matdes.2012.05.019

[39]M Bernini, R Hellmuth, C Dunlop, et al. Recommendations for finite element modelling of nickel-titanium stents-Verification and validation activities. *Plos One*, 2023, 18(8): e0283492. https://doi.org/10.1371/journal.pone.0283492

[40] Y Shao, J Yan, Y Zhi, et al. A universal packaging substrate for mechanically stable assembly of stretchable electronics. *Nature Communications*, 2024, 15(1). https://doi.org/10.1038/s41467-024-50494-8

[41] X Hu, S Li, P Peng, et al. Prosthetic heart valves for transcatheter aortic valve replacement. *Bmemat*, 2023, 1(2): e12026. https://doi.org/https://doi.org/10.1002/bmm2.12026

[42] C I Kwon, J P Moon,H Yun, et al. Evaluation of valve function in antireflux biliary metal stents. *BMC Gastroenterology*, 2018, 18. https://doi.org/10.1186/s12876-018-0878-8

[43]T F Yap, A Rajappan, M D Bell, et al. Thermally accelerated curing of platinum-catalyzed elastomers. *Cell Reports Physical Science*, 2024, 5(3): 101849. https://doi.org/https://doi.org/10.1016/j.xcrp..101849

[44]H M A Abdalla, D Boussaa, R Sburlati, et al. On the best volume fraction distributions for functionally graded cylinders, spheres and disks – A pseudospectral approach. *Composite Structures*, 2023, 311(5): 116784. https://doi.org/https://doi.org/10.1016/j.compstruct.2023.116784

[45] M Nemat-Alla, K I E Ahmed, I Hassab-Allah. Elastic-plastic analysis of two-dimensional functionally graded materials under thermal loading. *International Journal of Solids And Structures*, 2009, 46(14–15): 2774–2786. https://doi.org/10.1016/j.ijsolstr.2009.03.008

[46] A Ozturk, M Gulgec. Elastic-plastic stress analysis in a long functionally graded solid cylinder with fixed ends subjected to uniform heat generation. *International Journal of Engineering Science*, 2011, 49(10): 1047–1061. https://doi.org/10.1016/j.ijengsci.2011.06.001

[47]J H Kim, G H Paulino. Isoparametric graded finite elements for nonhomogeneous isotropic and orthotropic materials. *Journal of Applied Mechanics-transactions of the ASME*, 2002, *69*(4): 502–514. https://doi.org/10.1115/1.1467094

[48] K M Liew, S Kitipornchai, X Z Zhang, et al. Analysis of the thermal stress behaviour of functionally graded hollow circular cylinders. *International Journal of Solids And Structures*, 2003, 40(10): 2355–2380. https://doi.org/10.1016/S0020-7683(03)00061-1

[49]C T Loy, K Y Lam, J Reddy, Vibration of functionally graded cylindrical shells. *International Journal of Mechanical Sciences*, 1999, 41, 309–324. https://api.semanticscholar.org/CorpusID:136544318

[50]S HuiShen. Modeling and analysis of functionally graded carbon nanotube reinforced composite structures: A review. *Advances in Mechanics*, 2016, *46*(1): 478–505.

[51] P G Pauck, B D Reddy, Computational analysis of the radial mechanical performance of PLLA coronary artery stents. *Medical Engineering & Physics*, 2015, 37(1): 7–12. https://doi.org/10.1016/j.medengphy.2014.09.014

[52]M B Andersen, H Gregersen, K Kraglund, et al. Radial analysis of duodenal motility recordings in humans. *Scandinavian Journal of Gastroenterology*, 1991, 26(8), 843–851. https://doi.org/10.3109/00365529109037021

[53] A Kuchumov. Biomechanical modelling of bile flow in the biliary system. In V. M. Vassilev, S. G. Nikolov, M. D. Datcheva, & Y. P. Ivanova (Eds.), *13th National Congress on Theoretical and Applied Mechanics (NCTAM 2017),* 2018, 145. https://doi.org/10.1051/matecconf/201814504004

[54] Q W Leong, M LShen, K W Au, et al. A prospective, randomized study of the patency period of the plastic antireflux biliary stent: an interim analysis.

Journal Pre-proof

Gastrointestinal endoscopy, 2016, 83(2): 387–393. https://doi.org/10.1016/j.gie.2015.04.027

[55] T Hamada, H Isayama, Y Nakai, et al. Antireflux covered metal stent for nonresectable distal malignant biliary obstruction: Multicenter randomized controlled trial. *Digestive endoscopy*, 2019, 31(5): 566–574. https://doi.org/10.1111/den.13381

Journal Pre-proof